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Full Length Article

A novel decision support system based on computational intelligence and machine learning: Towards zero-defect manufacturing in injection molding

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ABSTRACT

Real-time monitoring solutions have gained popularity across industries due to the advent of Industry 4.0, AI, and big data enhancing the efficiency of industrial production and equipment decisions. Machine learning models that possess computing intelligence and interpretability provide superior predictive capabilities compared to manual adjustments, resulting in cost savings and manufacturing high-quality products. This study proposes a zero-defect manufacturing decision support system based on computational intelligence feature selection combined with interpretable machine learning. The decision support system integrates Particle Swarm Optimization (PSO) and the C4.5 decision tree method, abbreviated as PSO+C4.5, to enable the continuous monitoring of the injection molding process in real-time, considering production parameter information and collected data quality, guiding the decision-making process for implementing zero-defect manufacturing (ZDM). In contrast to existing research, our innovative methodology relies on computational intelligence techniques for extracting features and employs interpretable machine learning prediction models. In terms of quality prediction, our empirical findings show that the suggested method accomplishes the optimal balance between interpretability and predictive performance (Accuracy: 0.9889, Sensitivity: 0.9869, and Specificity: 0.9935). These characteristics can directly support maintenance personnel and operators in optimizing the processing quality process.

1. Introduction

As enterprises pursue higher profits and meet growing customer demands, the speed of producing new products has surpassed previous levels. There has been an increase in the difficulty of using system approaches for real-time monitoring and problem prevention in manufacturing [1] and rendering previously successful strategies ineffective or inefficient [2]. Furthermore, as production shifts from mass production to small-batch customized products [3]. There has been an increasing demand for customization, and a transition to lean manufacturing methods [4,5] to align customer demands. Consequently, the time for optimizing production processes has significantly reduced, increasing the proportion of defective products. Considering these reasons, it is essential to enhance and update more complex approaches and instruments [6,7] has become more important. There exists a pressing necessity to enhance quality management methodologies to effectively address current demands [8].

In the present business landscape, maintaining a competitive edge for businesses hinges on the production of top-quality goods and the reduction of waste [9]. In the manufacturing sector, subpar product quality can result in both direct and indirect costs, including repairs, transportation, and product returns, as well as sales losses due to customer dissatisfaction [10,11]. The underlying causes of manufacturing defects can be attributed to various factors, such as machine failure, tool wear, low-quality materials, or process variations caused by human errors [12]. Furthermore, Industry 4.0 technologies can effectively predict and rectify these process variations, thereby eradicating the root causes of defects. Achieving zero-defect manufacturing (ZDM) through the integration of Industry 4.0 technologies is a major objective in today's manufacturing industry [13]. If companies can achieve this objective, product quality and production efficiency will significantly improve, thereby granting them a competitive edge in the face of intense market competition.

ZDM is currently regarded as one of the most promising techniques [6,14], aiming to eliminate failures in the manufacturing process and achieve "getting it right the first time" [2], meaning eliminating defective parts during the production process. ZDM first appeared in internal US military documents during the Cold War era regarding defective

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weapon systems [15]. ZDM is a disruptive ideology and mindset that can fundamentally transform manufacturing.

There are two distinct methods to implement ZDM: process-oriented ZDM and product-oriented ZDM. ZDM with a product orientation focuses on resolving issues with the physical components themselves, whereas ZDM with a process orientation examines flaws in the manufacturing equipment and assesses the conformity of the finished goods accordingly. Taking into consideration the idea of predictive maintenance is the second component.

Instant monitoring technology can detect anomalies in real-time during the production process. Through the continual monitoring of equipment's operational status, it becomes possible to identify potential reasons for failures and implement timely remedial measures [16], making the ZDM concept easier to achieve. The emergence of Industry 4.0, artificial intelligence, and big data has garnered significant interest in real-time monitoring technology across diverse sectors. Employing data gathering and analysis, it is possible to optimize industrial output and enhance decision-making processes about equipment status [17]. Providing accurate insights into the state of production equipment enables the ZDM concept through new technologies that produce high-quality products by adjusting production parameters. ZDM includes four strategies: detection, repair, prediction, and prevention [18, 19].

The establishment of an integrated quality control system depends on the synergistic application of detection, repair, prediction, and prevention strategies. Upon identifying product defects, corrective actions can be implemented, while the collected data can be processed by specialized algorithm models to predict potential defect occurrences, allowing for proactive preventive measures. The lifecycle of a product or process typically encompasses various stages, such as raw material conversion, manufacturing, and assembly, during which product and process defects are more likely to emerge. When these defects arise early in the process, they can escalate progressively during subsequent stages, resulting in manufacturing issues and severe failures. Although some manufacturing defects may be inherently unavoidable due to operational reasons, a single defect can also stem from the accumulation of minor defects in preceding process steps. If these defects are not addressed promptly, they will persist and propagate across subsequent usage stages. Usage stage defects can also result from errors or improper instructions provided by the manufacturer. However, by gathering data on materials, processes, tools, and equipment and implementing quality control strategies [20], it is possible to significantly reduce various types of defects and adverse situations. Implementing comprehensive quality control measures can effectively address many manufacturing challenges.

Machine production parameters involve complex and non-linear relationships between machine components, products, or equipment [21]. In order to facilitate ZDM when confronted with such highly nonlinear and complex production issues, artificial intelligence is a crucial technology [9,22]. Integrating artificial intelligence into ZDM is a complex endeavor that entails various challenges, such as the need to gather data from production systems, a task that often demands customized approaches and consumes a significant amount of time [23]. Developing artificial intelligence solutions for ZDM necessitates a variety of competencies; domain expertise from production specialists is essential in addition to artificial intelligence techniques [22]. Previous research reports also pointed out deficiencies in data quality, including insufficient data or imbalanced data [24–26], and indicated a lack of interaction between ZDM and industry, with more research being conducted in laboratories rather than in factories [27].

This study aims to use artificial intelligence in the field of injection molding industrial manufacturing, with a specific focus on the detect-prevent approach inside ZDM. The decision support system integrates the PSO algorithm with the C4.5 decision tree method to continuously monitor the injection molding process in real-time. The decision-making process for ZDM deployment is informed by considering production parameter information and the quality of obtained data.

This paper provides three main contributions. First, in comparison to prior studies, our innovative methodology relies on an intelligent computing feature extraction strategy and an interpretable machine learning prediction model, a very uncommon approach in earlier research on ZDM. Second, when compared to other innovative machine learning (ML) models, our approach has been proven to provide the most optimal balance between predictive performance. Third, a decision support system for ZDM is constructed utilizing an interpretable machine learning model. The attribute of interpretability can offer scientific guidance and directly assist maintenance personnel and operators in optimizing the processing quality procedure. More precisely, these benefits could affect the enhancement of maintenance planning optimization and empower producers to decrease service costs through the augmentation of productivity and the maximization of downtime, thus attaining zero-defect manufacturing.

The remaining structure of this study is as follows: The next section describes the current state of ZDM, the rationale for using AI to improve manufacturing quality, and the methods used in this research. Section 3 describes the data and methods of the study and introduces the research domain. Section 4 presents the empirical results and their experiment. Section 5 discusses the results and explains their implications. Finally, conclusions are drawn.

2. Materials and methods

2.1. Zero defect manufacturing (ZDM)

The practice of ZDM is gaining greater attention as a result of the improvements in Industry 4.0 and smart manufacturing technologies [9, 22]. Phillip B. Crosby first introduced this concept while working as the quality manager for the Pershing missile program. He highlighted that the cost of quality may be eliminated by preventive measures rather than through evaluation, and he asserted that achieving zero faults is the benchmark for performance, not only an acceptable level of quality [28]. Although ZDM has only recently garnered extensive attention, its implementation has been facilitated by intelligent manufacturing and data-driven innovation. If there is sufficient data quantity and quality, and equipped with appropriate tools such as artificial intelligence, zero-defect manufacturing can be fully realized [19]. Utilizing real-time data to assess defect risks, ZDM helps to predict and eliminate potential defects promptly. For establishing a sustainable system, achieving early detection is crucial. AI plays a crucial role in advancing the ZDM concept [9,22].

Four fundamental basic strategies make up ZDM: detection (whether it be physical or virtual), repair, prediction, and prevention measures [27]. The detection strategy identifies defects, anomalies, and faults through analysis, classification, and identification based on parameters that may have adverse effects [22]. Although our research lies in physical detection, there is a growing interest in virtual detection as a result of enhanced data accessibility and decreased operational costs [29,30]. The most representative and extensively cited research in this field focuses on the detection capabilities of artificial intelligence. Using a segment-based deep learning framework, for example, to identify and categorize surface faults in the manufacture of commutators [31] and extended deep belief network method for chemical process fault diagnosis [32].

The prediction strategy forecasts flaws, abnormalities, and faults to estimate product component quality before manufacture using particular methods and historical data [22]. These models often include mathematical modeling and artificial intelligence techniques. One illustration of employing regression models is the utilization of these models to forecast two physical quality indicators within the tube extrusion process [33] and utilizing an extended deep belief network to maximize the utilization of valuable information and extract quality-related features from the batch production process [32].

The preventative strategy prioritizes the distribution of quality control and inspection equipment to oversee the performance of machines and the resulting quality of the related product [22]. It requires a thorough comprehension of procedures, doing machine condition analysis to detect possible flaws or deviations, and applying appropriate actions to mitigate issues [9]. The utilization of a deep decoupled convolutional neural network that requires mechanical interactions among composite components is referred to as intelligent composite fault diagnostic [34]. This method combines convolutional neural networks and extreme learning machines to create a new approach for fault diagnosis. Its purpose is to enhance the performance of fault classification and accelerate the learning process [35].

The repair strategy focuses on the refurbishment or remanufacturing of products within the sustainable supply chain [22]. In 2019, Gautam put forth a defect management model that took into account various factors, including carbon emissions [36], and Li et al. [37] put forth a methodology that entails the application of precision machining techniques, wire and arc additive manufacturing, and defect monitoring to correct metal defects.

According to Table 1, the Detection strategy is the most researched hotspot. This paper focuses on the Detection-Prevention strategy of ZDM. From Table 1, we know that when discussing individual strategies, the Detection strategy currently occupies the largest proportion (60%), while the research on the Prevention strategy is relatively low (9%). Simultaneously exploring the integrated research on the Detection-Prevention strategy is currently rare.

Furthermore, based on past research, most ZDM studies have used a single AI model, and few have used intelligent computing and interpretable machine learning models to adjust production parameters to produce high-quality products.

The first step is to collect machine production parameters (temperature, pressure, speed, etc.) from the production site, focusing on adjusting machine parameters to produce high-quality products and achieve ZDM. Therefore, Section 2.2 will introduce AI in artificial intelligence and the challenges commonly encountered in applying ZDM.

2.2. Artificial intelligence as an enabler for ZDM

The application of AI in ZDM typically includes sensors for perceiving the environment, AI techniques for analyzing data and making automated decisions, etc. AI techniques can learn from data to explore patterns of system behavior. For product examination, these data-driven approaches may employ conventional measurement systems or cameras. Supporting decision systems requires the incorporation of data and AI algorithms, and a variety of AI techniques can be applied flexibly based on the task at hand.

AI is infrequently incorporated into the operational models, strategies, production processes, and architectures of the majority of businesses nowadays [38,39], indicating that most enterprises fail to recognize how artificial intelligence technologies can improve production efficiency or lack relevant AI knowledge, thus not incorporating them into their production systems. The following will discuss three common challenges faced by the manufacturing industry when applying artificial intelligence to improve product quality.

Table 1
The distribution of published articles is a ZDM strategies.

ZDM strategies	Published articles distribution*
Detection	60%
Prediction	24%
Prevention	9%
Repair	4%
Defect mitigation or compensation	3%

* Average distribution result of the ZDM literature reviews ([9,22,27]).

2.2.1. Data availability and quality

The rapid development of industry 4.0 and the subsequent decline in sensor costs have substantially strengthened the volume of accessible data. In addition, the intricacy, dynamism, and multidimensionality of data sources have escalated [39]. The process of feature selection holds significant importance in data science projects as it serves to integrate unprocessed data and eliminate unnecessary details, resulting in the acquisition of data of superior quality. Which in turn, enhances the effectiveness and efficacy of artificial intelligence algorithms. Although the procedure is challenging and time-intensive, it does not yield immediate outcomes [40], it efficiently searches the feature space and finds the optimal subset of features, where optimized feature selection can enhance the performance of producing good products [41].

2.2.2. Imbalanced dataset

To ensure that AI models are void of bias, it is crucial in many industrial outputs to have a balance of defective samples to qualified products [42]. However, data imbalance is a prevalent issue when developing predictive quality models; various approaches, such as undersampling the majority class or oversampling the minority class, can be taken to address the situation [25]. Nevertheless, these methods alter the distribution of the solution space, which may reduce the predictive and generalization capabilities of the AI model.

2.3. Feature selection

The most crucial aspect is to identify key factors effectively and accurately from highly non-linear relationships and noisy data, which is a challenging task [21]. There is a strong interdependence among variables, making feature selection and the design of corresponding classifiers a critical problem in ZDM. The production environment is complex, and variables like noise make it challenging to pick out the most relevant attributes that capture the differences between classifications.

Achieving highly complex feature selection to enhance the classification performance of the system is crucial. Utilizing feature selection techniques to reduce dimensionality is very important.

2.3.1. Feature selection

Dimensionality reduction of feature information can be achieved through two main operations: feature selection and feature extraction. Feature selection mainly entails picking a set of features from a pool of D features, where the chosen subset contains d ($D > d$) features that can accurately represent the state of the system. In the case where both parameters are known, the total number of possible combinations is given by Eq. (1):

$$Q = C_D^d = \frac{D!}{(D-d)! \times d!} \tag{1}$$

If both D and d are sufficiently large, this can lead to an astounding number of combinations. Employing several metrics to assess the efficiency of feature selection methods leads to a computationally demanding procedure. In practice, the value of d is often unknown, further adding to the complexity of the calculations. Research into a fast and effective feature selection method for predictive maintenance and diagnostics has recently exploded in popularity, and it ranks high among AI's most pressing theoretical concerns [43,44].

2.3.2. Search strategy

Search strategies for feature selection fall into two main categories: exhaustive search strategy, and heuristic random search strategy.

(1) Exhaustive search strategy

Exhaustive search strategies include exhaustive algorithms and branch-and-bound algorithms. This approach guarantees finding the

optimal subset, the algorithms are very complex, requiring computational time complexity of the order of $O(2^n)$.

(2) Heuristic random search strategies

Heuristic random search strategies utilize heuristic algorithms to perform searches within the solution space. They encode the selection status of various types of feature information and utilize a classifier (fitness function) to evaluate the classification performance. Through iterative operations, these strategies aim to identify an approximately optimal subset. The attribute selection employment is implemented using the PSO algorithm, and C4.5 method as the fitness function.

2.4. PSO

"Particles" are the individual solutions to optimization problems in the search space that form the basis of the PSO method. A swarm is the technical term for a collection of particles. The fitness function determines the fitness level for each particle and gives each particle a velocity that controls the direction and amplitude of its displacement. The particles then move toward the current best particle in the search space. Assuming a solution space of D dimensions, the population of particles can be represented in a d-dimensional solution space with $X_i = [x_{i1}, x_{i2}, \ldots, x_{iD}]$, the flight velocity of particle i is represented as $V_i = [v_{i1}, v_{i2}, \ldots, v_{iD}]$, The best position in the solution space where each particle generates its best fitness value is $PB_i = [PB_{i1}, PB_{i2}, \ldots, PB_{iD}]$, The best position explored by the entire swarm up to the current moment is $GB_i = [GB_{i1}, GB_{i2}, \ldots, GB_{iD}]$. The particle swarm algorithm incorporates three parameters that influence the movement vector of a particle: (1) the particle's own inertia, (2) current state of the best particle solution, and (3) the state of the best overall solution across the cluster, as shown in Eq. (2):

$$v_{id}^{new} = w \times v_{id}^{old} + c_1 r_1 \left(pb_{id}^{old} - x_{id}^{old}\right) + c_2 r_2 \left(gb_d^{old} - x_{id}^{old}\right), \quad d = 1, 2, \ldots, D \tag{2}$$

In this case, c1 stands for cognitive learning and c2 for social learning. Both of these variables are constants that are positive and often between zero and four. Particles are controlled to converge by a slowing of their speed caused by the inertia weight w. The factor w typically falls in the range of 0.4 to 0.9. The variances r1 and r2 are evenly divided within the limits of 0 and 1. The particles' velocities are restricted within $[v_{\min}, v_{\max}]$. Furthermore, the velocity of the particle is constrained within the range of

$$x_{id}^{new} = x_{id}^{new} + x_{id}^{old}, \; d = 1, 2, \ldots, D; \; i = 1, 2, \ldots, N \tag{3}$$

The size of the swarm is represented by N. Fig. 1 illustrates the flow diagram of the PSO approach.

2.5. Decision tree

The C4.5 approach of the DT framework is employed as the fitness function for the PSO. C4.5 uses the concept of gain ratio based on entropy to measure the importance of variables as discriminating attributes. The weighted sum of $\log_2(p_j)$ is used for variables with multiple outcomes; the weights of these variables are equal to the probabilities of the outcomes, resulting in the following formula.

$$Entropy(X) = \sum_{j=1}^{c} p_j log_2(p_j) \tag{4}$$

Where p_j represents the occurrence probability of class j, indicating the ratio of X that is attributed to class j. If the target attribute can have c possible values, entropy is a measure of impurity in the training sample set. Thus, we can define an effectiveness indicator for attributes in data classification. The chosen metric is knowledge gain, which quantifies the anticipated decrease in entropy resulting from the classification of samples according to a specific property. In cases where the classification of the experimental set aligns with the expected classification of patterns, we posit that the prediction is accurate. The fitness function is technically characterized as the metric for evaluating the precision of classified elements.

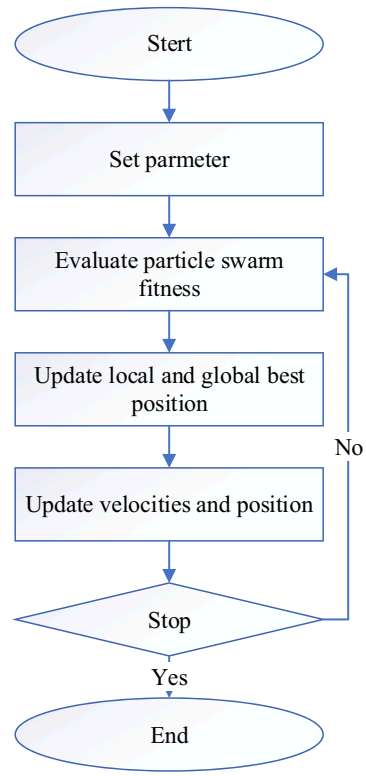

Fig. 1. The PSO process.

$$Accuracy = \frac{ofcorrectlypredictedexamples}{ofexamples} \tag{5}$$

Throughout the present study, the determination of fitness value is accomplished by the evaluation of classification accuracy, which is measured by the rate of recognition of the C4.5 classifier.

2.6. Indicators for performance

In binary classification systems, a threshold is defined to differentiate between the two groups. For this categorization issue, the confusion matrix summarizes the results of the predictions. As indicated in Table 2, the present study includes a confusion matrix to measure the model's performance and analyze the classification process.

The symbols TP, TN, FP, and FN, which stand for true positives, true negatives, false positives, and false negatives, respectively, are used to symbolize the various types of results. A confusion matrix is a common way to display the data. Our study includes more instances of high-quality products than low-quality ones. Thus, "Good" is given to items that are good and "Bad" to those that are bad. In this study, four performance metrics: Accuracy, Sensitivity, Specificity, and G-mean is determined from the values of TP, TN, FP, and FN (Eqs. (6) to 9).

Table 2
The confusion matrix.

		Predicted class	
		Good	Bad
Actual class	Good	TP	FN
	Bad	FP	TN

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{7}$$

$$Specificity = \frac{TN}{TN + FP} \tag{8}$$

$$G - mean = \sqrt{Sensitivity \times Specificity} \tag{9}$$

2.7. Ten-fold cross-validation

Examining the efficacy of the suggested model is the driving force for our study. Each subset of the total dataset is assigned a specific purpose: either training or testing. By utilizing 10-fold cross-validation, the current investigation mitigates the potential for bias introduced by the random selection of the training and testing sets [45]. Divide the dataset into ten non-overlapping subsets, with each subset representing around 10% of the whole dataset, and then run the model through 10-fold cross-validation. Then, nine subsets are created from the training set, and the tenth is used as the testing set. To determine the classifier's accuracy, we test it on each of the ten subsets independently before combining them.

2.8. Method for benchmarking

To determine the efficacy of the suggested methodology, we compare it to five existing classification techniques: Radial Basis Functions (RBF), C4.5 Decision Tree, Back Propagation Neural Network (BPNN), Support Vector Machine (SVM), and Self-Organizing Map. In the late 1980s, RBF was first used as a part of a two-layer neural network. Each covered unit was in charge of creating a radial-basis activation function. There is nonlinearity in the input to the RBF network, but uniformity in the output [46,47]. C4.5 is a classifier that uses a decision tree framework, proposed by Ross Quinlan [48] and commonly recognized as a statistical classifier. BPNN is a widely employed training procedure for artificial neural networks. The initial idea can be attributed to Paul Werbos during the 1970s. Multilayer neural networks are trained to learn, allowing them to discover any mapping between inputs and outputs [49]. The SVM algorithm is an effective ML technique that makes use of Kernels. SVM maps can transform non-linear input data into a feature space of higher dimensionality. In this transformed space, a linear decision boundary is constructed to categorize [50]. SOM uses unsupervised learning to construct a training sample input space that is low-dimensional and discretized, referred to as a map. In contrast to other ANNs, it employs a proximity parameter as a way to represent the topographical aspects of the area of input. The initial exposition of the model was provided by Teuvo Kohonen in the form of an ANN, and it is occasionally called a Kohonen map or network [51]. SOM are nonlinear extensions of Principal Component Analysis (PCA) that provide numerous benefits in comparison to traditional feature extraction methods [52,53].

3. Research methodology

3.1. Method presentation

The implementation of the PSO+C4.5 algorithm that we propose is represented in Fig. 4. The intelligent computing PSO model reduces noise in the data to enhance model performance and robustness. The explainable DT model increases the model's reliability and makes it easy for on-site personnel to use. The performance of the algorithm is assessed by the utilization of a 10-fold cross-validation methodology, as depicted in Fig. 2. The utilization of the 10-fold cross-validation technique guarantees a rigorous assessment of the algorithm's efficacy through iterative training and testing on distinct subsets of the dataset. This approach helps in assessing the algorithm's generalization capabilities and its performance across diverse data instances. Cross-validation mitigates biases and enhances the reliability of the algorithm.

Transfer Learning improves model learning efficiency and performance by transferring and applying knowledge from one domain or task to another similar domain or activity. Applying Transfer Learning to the PSO algorithm offers the following advantages:

1. Accelerating convergence speed by reducing the iterations required to explore the solution space.
2. Enhancing algorithm performance, leading to an improved likelihood of finding the global optimum solution, thereby improving the algorithm's performance.
3. Reducing computational costs by lowering computational complexity and conserving computational resources.
4. Enhancing algorithm adaptability: by adapting to different optimization problems, thereby enhancing the algorithm's generality and applicability.

In the literature, several studies have explored the integration of Transfer Learning with PSO. For instance, Zhan et al. [54] proposed a knowledge transfer-based PSO algorithm, which accelerates the convergence speed of PSO by acquiring experiential knowledge from a solved problem and transferring it to a new problem.

This study uses Transfer Learning and sets the crucial parameters according to Wang et al. [14]: the cognitive learning factor (c1) and social learning factor (c2) are both adjusted to 2, and the particle velocity lower and upper limits (v_{min} and v_{max}) are set to minus 4 and 4, respectively. The dataset has as many particles as features. Inertia (w) is 0.4. Iteration continues until the particle frequency hits 100% or the maximum cycle number (T). For the reason of consistency, T was fixed to 150. Additionally, each algorithm is executed 30 times with a variety

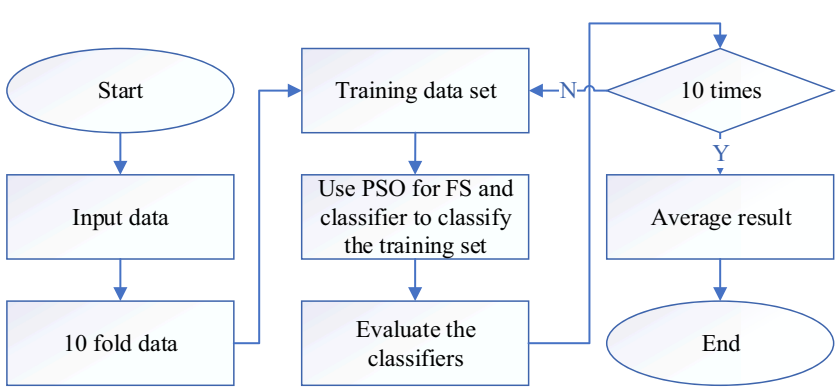

Fig. 2. The implementation of the proposed algorithm.

of random seed values, and the average of the best solution is collected. By configuring these parameters and repeating the algorithm with various random seed values, the algorithm's performance is comprehensively assessed, and the stability of the results is ensured across multiple runs.

3.2. Case study

The production sector relies on injection molding, a processing procedure, to create plastic goods with complex geometries and a high degree of precision [55,56]. Injection molding high-value objects has substantially influenced various sectors [57], including food containers, medical micro-components, intricate structural parts for the electronics industry, and applications in the optical industry, among others. The inception of injection molding technology can be attributed to John Wesley Hyatt, who developed the first injection molding equipment in 1872. In 1956, William H. Willert presented the screw-type injection molding machine, which was a turning point in the evolution of modern injection molding machines [58].

Fig. 3 illustrates the operational mechanism of an injection molding machine. The generation of melt-pressure occurs through the rotation of the screw, which compels the polymer melt to enter the cavity of the mold. The melt pressure is partitioned among three phases or sub-phases of injection within the mold cavity: (a) initial filling, (b) consistency filling, and (c) compressive melting. The melt resistance in the runner directly influences the melt pressure that occurs during the initial filling process. Enhancing the initial melt pressure or reducing the first melt temperature typically leads to an acceleration in melt fluidity. The molten material that passes through the runner must be at a higher pressure to overcome the resistance. Melt pressure in the consistent filling is determined by mold cavity permeability. Higher mold temperature improves melt fluidity, lowering permeability and melt pressure. Compressive melting is contingent upon the level of friction of compressing the polymer melt. In this sub-stage, it is observed that the melt pressure rises rapidly, ultimately reaching its maximum level during the injection phase. Hence, during the manufacturing process, variables including pressure, volume, viscosity, injection force, plasticization time, and injection time are intricately linked to the predictive maintenance of injection molding machines.

The injection molding machine is equipped with various sensors, such as temperature and pressure sensors. These sensors provide readings that offer real-time data about the clamping unit, injection unit, and mold behavior during the injection molding process. This real-time information allows for a focus on ZDM through detection and prevention. Table 3 lists the variables included in the dataset. x1 represents Total Production Rejections (Rejects cause): It refers to the discrepancy between the production parameters set before actual production (injection cycle, computer image analysis, quality program, mold temperature, etc.) and the actual parameters during production. When this discrepancy is significant, the system classifies the product as defective. Total Production Rejections correspond to the defective items produced due to

Table 3
Factors.

No.	Items	Descriptive
x1	ASZ	Rejects cause
x2	SZx	Shot counter actual value
x3	IQFitQual	Pressure profile compliance
x4	IQInjVol	Injection volume
x5	IQrViscChng	Change in viscosity
x6	ZSx	Injection time actual value
x7	ZDx	Plasticizing time actual value
x8	EA	Injection force
x9	ACPx	Material cushion smallest value
x10	ACPn	Material cushion smallest after holding pressure
x11	ASSx	Injection volume
x12	AC3u	Switch over volume
x13	ACPv	Material cushion end holding pressure
x14	Mm	Torque mean value current cycle
x15	APVs	Specific injection pressure peak value
x16	FZx	Flow number
x17	SKs	Clamping force peak value
x18	SKg	Clamping force stored
x19	ZUaxs	Standstill time prior to cycle start
x20	ZUs	Cycle time until end demolding
x21	H2x	Nozzle
x22	H3x	Barrel zone 1
x23	H4x	Barrel zone 2
x24	H5x	Barrel zone 3
x25	H8x	Feed throat

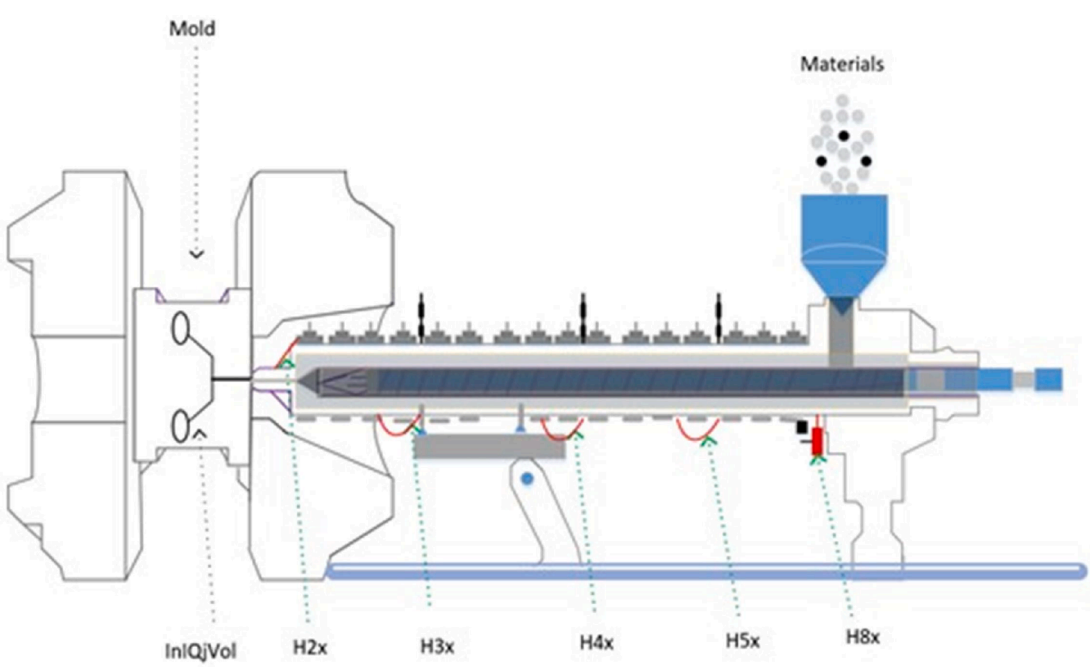

Fig. 3. The process of injection molding.

parameter differences in the production process. x2 stands for Actual Cycle Counter Value, which indicates the actual number of products produced. If it does not match the number of injection actions performed by the injection molding machine, it implies unfinished injection actions. x3 represents Pressure Profile Compliance. It signifies the graphical depiction of variations in adjusting injection material volume and measuring material viscosity. The absolute pressure during injection is denoted along the y-axis, while the screw stroke is represented along the x-axis. x4 is Injection Volume, denoting the actual value of screw injection volume. It detects changes in volume caused by differences in the closing time of the screw's check valve or the flow of molten material during screw retreat. x5 signifies Change in Viscosity. It illustrates the differences between the flow characteristics of molten material during production filling and reference values. Adjusting switch pressure values can compensate for changes in material properties. x6 stands for Actual Injection Time Value. It involves calculating the actual injection time by switching the position of the screw sensor, following alterations in settings. x7 represents the Actual Plasticization Time. It monitors the time difference between the previous cycle and the current cycle. x8 is Injection Force. It refers to the mechanical force required for the forward movement of the screw during injection and holding phases. x9 denotes Minimum Residual Material Quantity. This refers to the minimum amount of residual material displayed by the control unit at the front end of the screw. x10 represents the Residual Material Quantity after holding the pressure. During the concluding stage of the holding pressure phase, the quantity of material that is discharged under the influence of holding pressure is seen and measured using sensors. Table 3 presents the list of variables included in the dataset. x11 is Actual Injection Volume. It refers to the set value of screw injection volume. x12 is Switching Point Volume, which calculates the volume required for the screw's position to switch from injection to holding pressure termination. x13 is the Minimum Residual Material Quantity, used to indicate the minimum residual material quantity used throughout the entire holding pressure process. x14 is the Torque Mean Value Current Cycle. It denotes the torque during plasticization, with the value assisting in monitoring the material plasticization process. x15 stands for Specific Injection Pressure Peak Value. It represents the maximum absolute injection pressure during injection and holding pressure processes, which can be considered a safety setting for the mold. x16 is the Flow Number. It is monitored using ENGEL's Microplast computer melt index control program, which is the average injection pressure (flow number) during the dynamic injection filling phase. x17 signifies Clamping Force Peak Value. The clamping force setting opposes the injection pressure to maintain mold closure. Incorrect upper/lower limits of clamping force setting may lead to mold damage. x18 represents Stored Clamping Force, indicating the measured clamping force magnitude at the end of establishing clamping force. x19 is Pre-Cycle Pause Time: It signifies the time interval between the end of a production cycle and pressing the start button. x20 is Cycle Time Until Mold Release Termination. It indicates the total cycle time from initiating production in a cycle until the mold pause time concludes.

Temperature Control System: Starting from the nozzle closest to the mold (x21), barrel zone 1 (x22), barrel zone 2 (x23), and barrel zone 3 (x24), as well as the material inlet feed throat (x25). After material drying, the material is directly fed into the feeding port through the central feeding system. Therefore, it is necessary to monitor the temperature at the feeding port. x25 represents the temperature at the feed port (raw material inlet) and is controlled using a cooling water circuit, which is used to monitor the temperature at the feeding port. The temperature control of the barrel is divided into three zones: x24 representing Barrel Zone 3, x23 representing Barrel Zone 2, and x22 representing Barrel Zone 1. The nozzle x21 reaches the highest temperature to melt the material, which is then injected into the mold according to the set production parameter conditions.

3.3. Decision support system

The decision support system enables continuous monitoring of the injection molding process by integrating the PSO and C4.5 decision tree techniques, denoted as PSO+C4.5. This study referenced previous literature [59], and the architectural framework of the proposed Decision Support System (DSS) is presented in Fig. 4. The DSS comprises a total of 25 components, all of which are based on machine learning and the Internet of Things.

(1) Data Collection: Internet of Things (IoT) sensing technology relies on Message Queuing Telemetry Transport (MQTT). MQTT enables devices to efficiently transmit small data packets over the network, utilizing control messages as small as 2 bytes. It ensures messages are transmitted in a concise manner, minimizing network bandwidth and power consumption. A SQL database is used for data synchronization and collection prior to storing the data in Azure Blob cloud storage.
(2) Feature Selection: PSO is executed for feature selection.
(3) Prediction Model: In a Decision Tree (DT) model, the gathered processing parameters are used as predictive variables to estimate the state and perform data adjustments.
(4) Cloud Architecture: The ML model is deployed using Docker containers, which provide API endpoints for testing new data that has been extracted. The Azure ML service offers a cloud-based environment that enables the deployment and updating of ML models. ML model weights and prediction errors% are stored using Azure Blob storage.
(5) Data Analysis: A GUI-based interface displays ML model results for maintenance personnel and operators.

4. Experiments and results

4.1. Current process

We obtained actual production data from an injection molding factory. The factory produces various plastic products, including doormats, bottle caps, and bottle openers. For our study, we exclusively focused on the production of doormats, which are susceptible to manufacturing defects including Flash, Short Shot, and Warpage, all of which have an enduring effect on process stability. These doormats enhance the seal between the door lock and the steel door. The mold design adopts a symmetrical layout with four cavities to ensure production consistency. The product's appearance is depicted in Fig. 3.

In the experiment, we used a common-grade Low-Density Polyethylene (LDPE-M2100) as the polymer material. Before each production run, 50 gs of black masterbatch (LDPE-548) were mixed with 1 kg of LDPE-M2100 for 20 min. Then, the material mixture was placed in the drying hopper of the central feeding system for heating and dehumidification. The material was heated to 80 °C and maintained for 2 h to ensure its dryness. The production environment was maintained at a constant temperature of 25 °C. To ensure uniform filling of the material in the injection molding machine barrel, a 3-minute trial injection was performed before formal production runs to purge any inadequately plasticized material from the front end. This step aimed to ensure consistent manufacturing conditions for each test series. Additionally, achieving thermodynamic equilibrium conditions is crucial for ensuring the reproducibility of the experiment. Therefore, measurements were taken after producing the first ten parts to achieve thermal stability. All experiments were conducted on an ENGEL e-mac 75 electric precision injection molding machine. This machine is renowned for its superior efficiency, accuracy, and exceptional precision, ensuring the reproducibility of the experiments. Fig. 5 provides a three-dimensional representation of the doormat and product design.

Three categories of defect can be found in bad products:

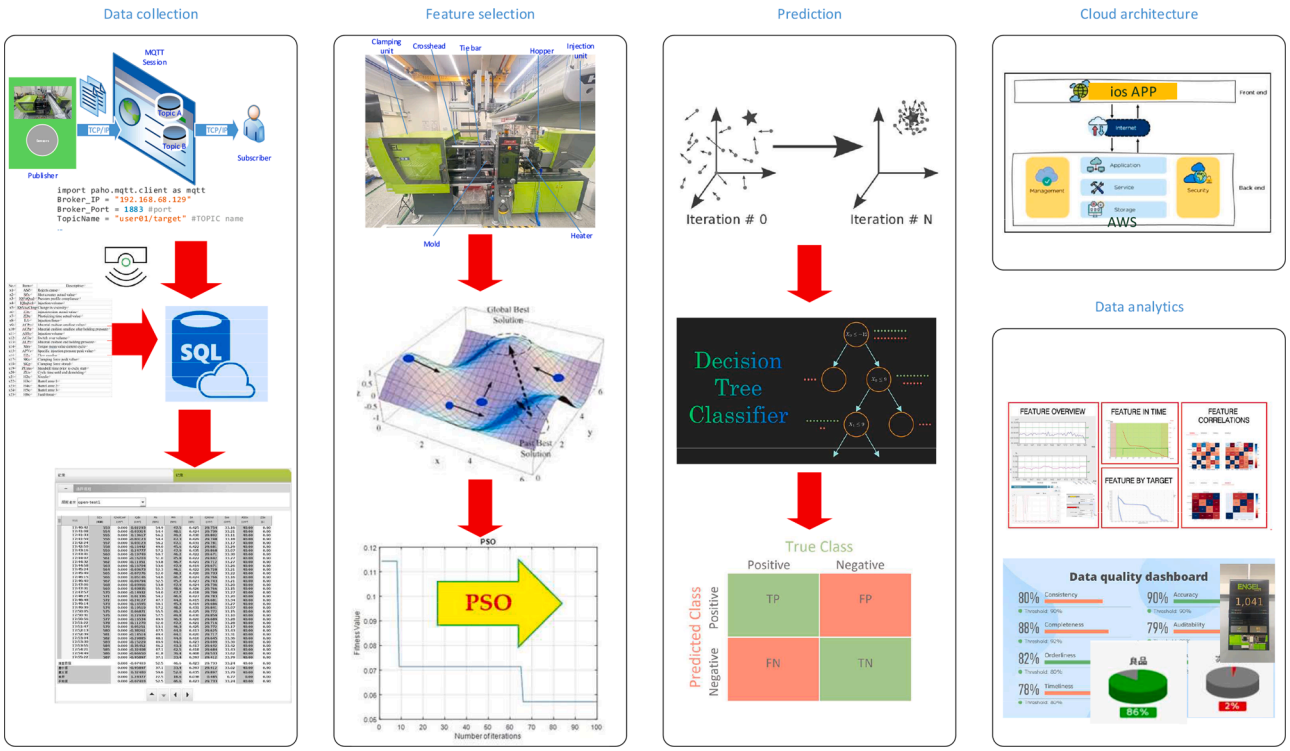

Fig. 4. Decision support system.

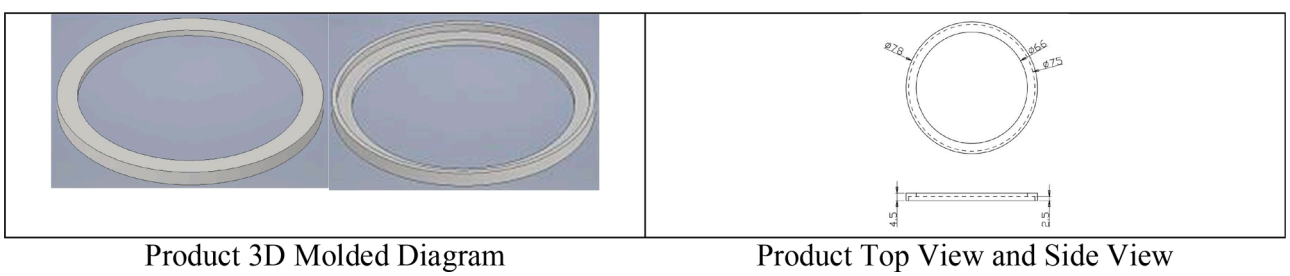

Fig. 5. 3D and product design drawing.

1. Flash: This problem usually arises when the product is not correctly ejected from the mold, producing complications in succeeding cycles. It contains surplus material beyond the product's form.
2. Short Shot: This defect results from inadequate material filling, leading to insufficient product weight. This defect is relatively rare for this particular product.
3. Warpage: Warpage is primarily caused by premature ejection of the product from the mold before it has fully cooled. This occurs when the ejection rod pushes the product out while it is still in a semi-molten state. This issue can be resolved through manual reworking to correct the product's shape.

Fig. 6 shows the inspection process for the product, distinguishing between defective and non-defective items.

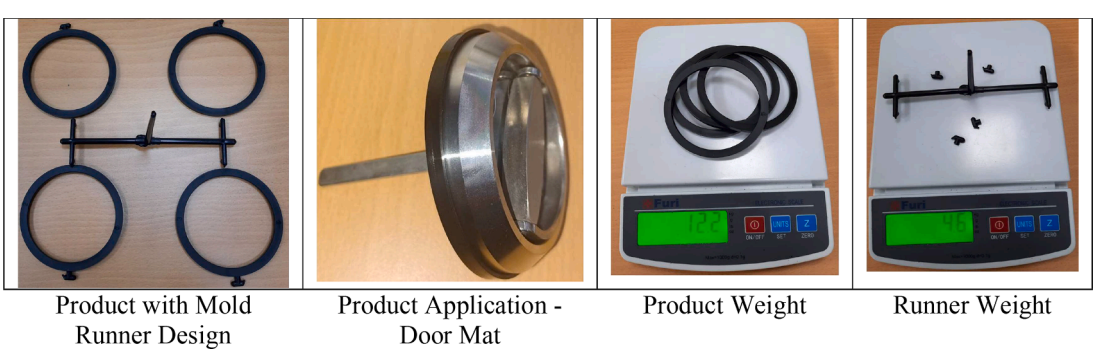

Fig. 6. Product Inspection.

4.2. Data preparation

The data were collected in 2022, and the product involved is a door lock gasket designed for metal doors. The initial dataset analysis in this study involves data pretreatment, which includes important procedures such as data cleaning, data integration, data transformation, and data dimensionality reduction. Starting from the initial 2100 records, after data preprocessing, the dataset was refined to contain 1988 production data points and 25 variables. The dataset was divided into two classes: the "good" class, containing 1370 records (68.91%), and the "defective" class, containing 618 records (31.09%). Table 4 provides the descriptive statistics.

4.3. Results

To comprehensively assess the performance of the proposed method in this paper, we conducted a comparative experiment to compare it with five common machine learning algorithms: BPNN, SVM, C4.5, SOM, and RBF. As illustrated in Table 5, the experimental findings indicate that our proposed PSO+C4.5 hybrid method, which integrates particle swarm optimization with the C4.5 decision tree, notably surpasses the other five benchmark algorithms in terms of classification accuracy. This outcome strongly confirms the superior performance of the proposed new method in executing classification prediction tasks, thereby highlighting its outstanding classification performance.

G-mean is the geometric mean of Sensitivity and Specificity. In this study, the PSO+C4.5 hybrid method performed excellently in terms of the G-mean indicator, indicating that the method demonstrates outstanding performance in both sensitivity and specificity. This implies it has extremely low rates of false negatives (Type I error) and false positives (Type II error). This result confirms the high accuracy and robustness of the proposed new method in classification prediction tasks, efficiently distinguishing between positive and negative samples and reducing the risk of detection errors.

The performance differences between various methods were assessed by creating a box plot in Fig. 7, which illustrates that the PSO+C4.5 hybrid method exhibits the highest average G-mean value while also having the smallest data distribution dispersion and the lowest variance. This demonstrates the reliability and stability of the method for classification tasks, and indirectly validates the effectiveness of the hybrid method design.

Table 4
Descriptive statistics.

No.	Items	Mean	S.D.	Min.	Max.
x1	ASZ	47.28	105.72	0.00	18,797.00
x2	SZx	235.67	156.80	1.00	1214.00
x3	IQFitQual	96.32	5.46	0.00	100.00
x4	IQInjVol	30.93	0.44	29.24	32.36
x5	IQrViscChng	3.64	2.76	−2.67	11.13
x6	ZSx	0.58	0.02	0.03	1.41
x7	ZDx	15.22	4.61	0.00	202.66
x8	EA	0.22	0.01	0.00	0.49
x9	ACPx	0.74	1.02	−0.02	30.13
x10	ACPn	0.78	1.07	0.00	31.40
x11	ASSx	34.88	0.19	31.99	84.82
x12	AC3u	4.99	0.17	0.00	5.00
x13	ACPv	0.72	0.24	−0.02	6.33
x14	Mm	25.92	1.26	0.00	32.30
x15	APVs	346.01	13.46	18.90	921.80
x16	FZx	0.00	0.00	0.00	0.00
x17	SKs	138.52	6.80	29.90	195.80
x18	SKg	132.18	11.88	27.60	144.00
x19	ZUaxs	17.01	429.97	0.00	12,677.82
x20	ZUs	53.66	431.88	0.00	12,771.84
x21	H2x	206.20	2.22	200.60	210.20
x22	H3x	193.22	2.23	189.50	200.50
x23	H4x	189.86	1.91	184.70	195.20
x24	H5x	180.90	2.41	172.10	189.60
x25	H8x	44.85	1.49	44.00	58.30

Table 5
Mean performances of all the models.

Model	G-mean	Sensitivity	Specificity	Accuracy
PSO+C4.5	0.9902	0.9869	0.9935	0.9889
SVM	0.8509	0.8255	0.8770	0.8415
BPNN	0.3114	0.9080	0.1068	0.6590
C4.5	0.4780	0.9350	0.2443	0.7203
RBF	0.9162	0.9715	0.8641	0.9381
SOM	0.9219	0.9745	0.8722	0.9427

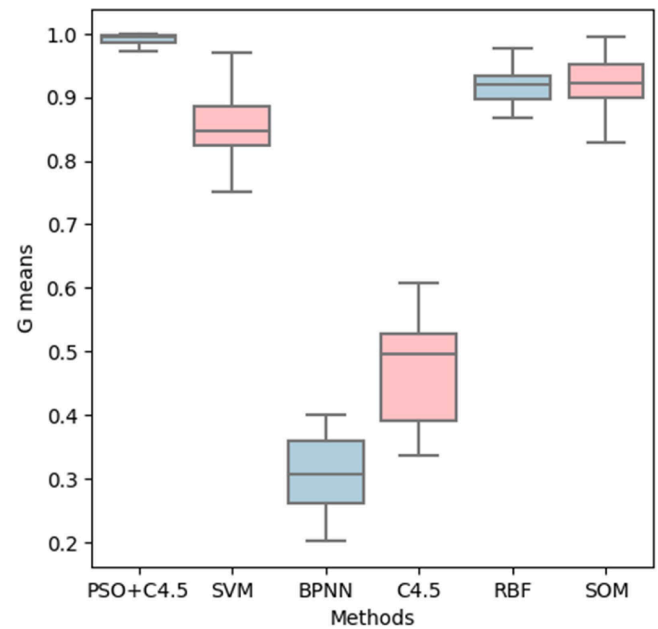

Fig. 7. Boxplots.

The box plot effectively illustrates the considerable disparities in the G-mean scores across the diverse methods, allowing for an unambiguous and direct comparison between them. This presents dependable empirical data that endorses the exceptional performance of the suggested technique. The box plot not only efficiently displays the data dispersion but also emphasizes the remarkable benefits of the PSO+C + 4.5 method. This offers a clear and persuasive experimental proof of the efficacy of this innovative approach.

Utilizing the Statistical Package for the Social Sciences (SPSS) software, we quantitatively evaluated the performance of multiple classification models. A one-way analysis of variance (ANOVA) was executed at a significance level of 0.05 to preliminarily investigate any significant discrepancies among the various models with respect to G-mean scores. The ANOVA results (Table 6) indicated the presence of statistically significant disparities among the algorithm models in terms of this performance metric.

Following the completion of the ANOVA test, we proceeded to conduct Tukey's Honestly Significant Difference (HSD) post hoc multiple comparison test to determine the statistical significance of the differences in G-mean scores among the algorithmic models. The detailed results of this analysis can be found in Table 7. The use of Tukey's HSD

Table 6
ANOVA for average G-means.

Source	Df.	SS	MS	F	*p*
Method (Factor)	5	11.757	2.351	907.034	0.000
Error	174	0.451	0.03		
Total	179	12.208			

Table 7
Caption.

	PSO+C4.5	SVM	BPNN	C4.5	RBF	SOM
PSO+C4.5		**0.000**	**0.000**	**0.000**	**0.000**	**0.000**
SVM			**0.000**	**0.000**	**0.000**	**0.000**
BPNN				**0.000**	**0.000**	**0.000**
C4.5					**0.000**	**0.000**
RBF						0.999
SOM						

test is particularly noteworthy as it provides a highly standardized approach to evaluating and comparing the performance metrics of various algorithmic models, including G-mean scores. The results indicate that the method presented in this study significantly outperforms other methods in terms of G-means.

5. Discussion

5.1. Discussion

According to the results of the experiments, we establish the following conclusions:

1. The G-mean metric indicates that the classifier that performs best is PSO+C4.5, followed by RBF and SOM as the second and third best-performing algorithms, respectively. Furthermore, accuracy, sensitivity, and specificity metrics all support PSO+C4.5 as the superior hybrid classifier.
2. The proposed PSO+C4.5 approach improves minority class classification accuracy in terms of sensitivity and specificity metrics, which is essential for detecting defective products. The PSO+C4.5 model achieves this by monitoring the actual state of the equipment, enabling the maintenance of expected production levels and reducing overall maintenance costs.
3. The proposed PSO+C4.5 algorithm selected eight important factors from the original 25 factors, namely x2, x3, x4, x5, x20, x21, x22, and x23. The reduced number of factors not only reduces setup costs by minimizing sensor usage but also alleviates the operational burden of sensor information. This enhances the feasibility of ZDM.

The PSO+C4.5 algorithm demonstrates excellent performance in terms of classification accuracy, sensitivity, and specificity. It provides valuable insights for maintaining production quality and reducing costs through effective monitoring and ZDM. Below, we will discuss in more detail the important factors selected by the PSO+C4.5 algorithm.

5.2. Factor discussion

Regarding predictive maintenance for injection molding, there are eight important factors to consider. These factors are the actual value of cycle counter (x2), consistency of pressure profile (x3), injection volume (x4), change in viscosity (x5), cycle time until mold release (x20), actual value of temperature zone (barrel zone) 2 (x21), actual value of temperature zone (barrel zone) 3 (x22), and actual value of temperature zone 4 (barrel zone) (x23).

The quality of plastic goods is highly influenced by various aspects, including the melt temperature in the barrel, injection speed, holding pressure, cooling time, and mold temperature, as indicated by numerous research findings. Increased melt temperature lowers viscosity and improves flow within a given range. This also contributes to increased product weight and enhances the consistency of product thickness across various regions within the mold cavity. The mold cavity volume remains constant, so raising the injection speed can lead to shear heating of the melt in the barrel, subsequently elevating the melt temperature. Variations in backflow and specific volume transpire as the melt temperature attains a specific threshold, resulting in a marginal diminution in both the thickness and weight of the product. An increase in holding pressure and maintaining it for a set duration raises the pressure in the mold cavity, thereby increasing the product weight (density) and thickness. Extending the cooling time affects product weight to a certain extent, but it also elongates the injection molding cycle time, potentially reducing production output [60–62]. As a result, as reported in the previous literature, this study considers the critical factors for injection molding to include consistency of pressure profile (x3), injection volume (x4), and change in viscosity (x5).

The remaining parameters, which were not highlighted in previous literature as important factors, include the actual value of cycle counter (x2), cycle time until mold release (x20), and the actual value of temperature zone (nozzle) (x21), the actual value of temperature zone (barrel zone) 1 (x22), and actual value of temperature zone 2 (barrel zone) (x23). The actual value of the cycle counter (x2) indicates the frequency of production interruptions caused by factors such as personnel scheduling, urgent orders, or material shortages. Despite using identical production conditions (mold, injection product, material, machine), an increased number of production interruptions introduces noise and instability into the production process. Cycle time until mold release (x20) refers to the total time until the mold is released before any pauses. The reasons for these pauses may stem from instances where the molded product does not detach from the mold. In such cases, the injection molding machine transitions from fully automatic to manual mode for product removal. Failure to release from the mold could arise from material, mold, or injection molding parameter issues. If material and injection molding parameters are confirmed to be correct, mold wear due to production can cause sticking. When adjusting injection molding parameters proves ineffective, maintenance of the mold might be necessary. Parameters x21, x22, and x23 correspond to the temperature control devices, encompassing the nozzle, barrel zone 1, and barrel zone 2. Temperature control in all of these zones significantly impacts product quality. The nozzle temperature, being closer to the mold, has a greater impact on product formation. Constant vigilance is necessary to inspect and adjust these temperature control devices. If the temperature control unit displays excessively high or low temperatures, the control unit should be examined. In the case of abnormal temperature readings in the barrel zones, predictive maintenance might involve replacing thermocouple sensors.

In summary, the impact of machine failures extends beyond immediate repair costs to include expenses related to maintenance, downtime, and spare parts. Implementing effective ZDM strategies, such as the proposed PSO+C4.5 algorithm, can reduce these losses by proactively identifying and addressing potential issues during the injection molding process.

Additionally, this study develops a ZDM decision support system capable of gathering and controlling pertinent process characteristics from the Injection Molding Machine (IMM). Real-time information, including time, temperature, and pressure, can be collected through the communication ports of the injection molding machine. Subsequently, these data can be transmitted to the management system to achieve real-time feedback on injection molding parameters. This real-time production data analysis can guide decision-making processes.

Incorporating the findings of research studies can yield numerous advantages. Firstly, the production process can be monitored remotely in real-time, enhancing process oversight. Secondly, injection molding machines can provide insights into production behavior. Given this consideration, it is feasible to implement preventive maintenance protocols on an injection molding machine. Engineers can monitor operational parameter variation in real-time, and when these parameters exceed set thresholds, immediate predictive maintenance can be initiated. Maintenance expenses, downtime, spare component requirements, and equipment failure financial losses are reduced using this strategy.

6. Conclusion

Our study proposes a ZDM decision support system based on intelligent computation feature selection combined with interpretable machine learning. The integration of PSO and the C4.5 decision tree method, referred to as PSO+C4.5, improves the capability of the decision support system to monitor the injection molding process in real-time.

We implemented the suggested ZDM decision support system framework in the domain of injection molding. Through a real manufacturing plant case study, we obtained actual data from a single injection molding production line in 2022. The product considered was a door lock gasket. After preprocessing, the original dataset was refined from 2100 entries to 1988 production data points and 25 variables. To ascertain the efficacy of the suggested methodology, a comparative analysis was conducted with five alternative approaches. SVM, BPNN, C4.5 decision tree, RBF, and SOM. The results indicate that the PSO+C4.5 proposed method achieved significantly better performance. The success of the PSO+C4.5 method provides empirical evidence of the effectiveness of the proposed preventive maintenance strategy in the specific domain of injection molding. This framework's ability to utilize real-time monitoring data for ZDM may have a significant impact on improving production efficiency and reducing costs in the injection molding process.

We recommend that future research should investigate the drawbacks of the PSO algorithm, including difficulty in escaping from local optima and premature convergence. To overcome these constraints, we suggest integrating a novel local search method, such as hybrid PSO techniques, to further augment the efficacy of the PSO algorithm.

Both the final product's quality and the production process's efficiency are affected by the production parameters that are established in the injection molding field. Our research centered on the door lock seal gasket, which possesses a mold design featuring four cavities and a symmetrical shape. In the future, extending this method to asymmetric products or using different molds would be advantageous. Additionally, including various parameters such as machine vibration or sound has the potential to further expand the possibilities of ZDM.

CRediT authorship contribution statement

Jiun-Shiung Lin: Supervision, Project administration, Data curation. **Kun-Huang Chen:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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